

Pseudo Code

1. User Authentication

```
BEGIN
DISPLAY 'Login Form'
INPUT username
INPUT password
IF username = valid AND password = valid THEN
    DISPLAY 'Login Successful'
    OPEN Main Menu
ELSE
    DISPLAY 'Invalid Username or Password'
END IF
END
```

2. Live Camera Monitoring

```
BEGIN
START Camera
WHILE Camera is Active DO
    CaptureFrame()
    DetectStudent()
    ExtractPoseKeypoints()
    SavePoseData()
END WHILE
END
```

3. Pose Validation

```
BEGIN
READ PoseData
IF PoseData is Valid THEN
    SaveValidatedPose()
ELSE
    DiscardPoseData()
END IF
END
```

4. Suspicious Behavior Detection

```
BEGIN
READ ValidatedPose
ExtractBehaviorFeatures()
Prediction ← DeepLearningModel(BehaviorFeatures)
IF Prediction = Suspicious THEN
    FlagBehavior()
    AnnotateStudent()
    SaveIncident()
ELSE
```

```
    MarkAsNormal()  
END IF  
END
```

5. Behavior Classification

```
BEGIN  
IF Behavior='Side Glancing' THEN RecordBehavior()  
ELSE IF Behavior='Hand Signals' THEN RecordBehavior()  
ELSE IF Behavior='Standing Up' THEN RecordBehavior()  
ELSE IF Behavior='Passing Notes' THEN RecordBehavior()  
ELSE IF Behavior='Head Down' THEN RecordBehavior()  
ELSE IF Behavior='Bounding Box Collision' THEN RecordBehavior()  
ELSE RecordNormalBehavior()  
END IF  
END
```

6. Alert Generation

```
BEGIN  
IF SuspiciousBehaviorDetected THEN  
    GenerateAlert()  
    DisplayAlert()  
    SaveToDatabase()  
ELSE  
    ContinueMonitoring()  
END IF  
END
```

7. Module Management

```
BEGIN  
DISPLAY ModuleMenu  
INPUT UserChoice  
CASE UserChoice OF  
    Add: SaveModule()  
    Edit: UpdateModule()  
    Delete: DeleteModule()  
END CASE  
END
```

8. Data Retrieval

```
BEGIN  
DISPLAY DataMenu  
INPUT Selection  
IF Selection='P1' THEN RetrieveP1Data()  
ELSE IF Selection='P2' THEN RetrieveP2Data()  
END IF  
DisplayResults()  
END
```

9. Overall System Process

```
BEGIN
Login()
IF LoginSuccessful THEN
    StartCamera()
    WHILE ExaminationIsRunning DO
        CaptureFrame()
        DetectStudent()
        ExtractPoseKeypoints()
        ValidatePose()
        ExtractBehaviorFeatures()
        Prediction ← DeepLearningModel()
        IF Prediction = Suspicious THEN
            FlagBehavior()
            AnnotateStudent()
            GenerateAlert()
            SaveIncident()
        ELSE
            ContinueMonitoring()
        END IF
    END WHILE
ELSE
    DISPLAY 'Access Denied'
END IF
END
```